

Defusing Affective Polarization with a Co-Authored Community Note: A Field Experiment with Rival K-pop Fandoms

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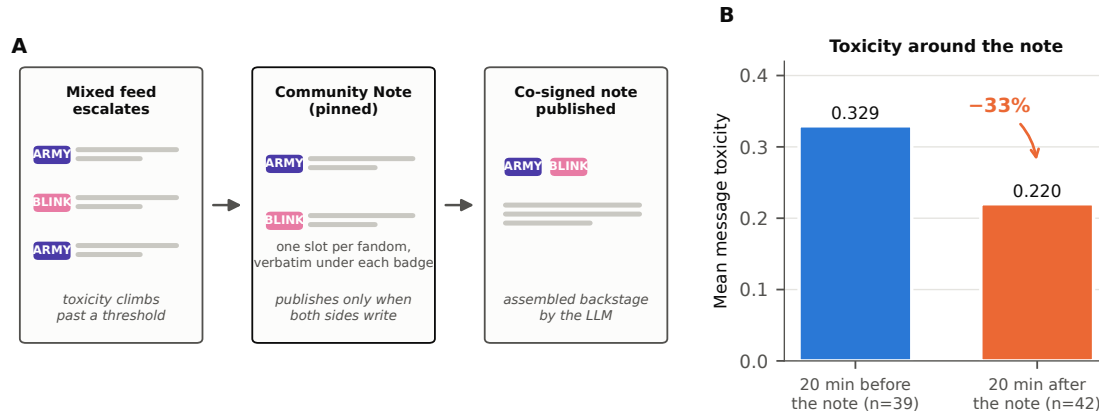


Fig. 1. The Community Notes intervention and its headline effect. (A) When toxicity in a mixed ARMY-BLINK feed crosses a threshold, the platform pins a Community Note with one contribution slot per fandom; the note publishes only when both sides have written their part, each entry verbatim under its author's badge, assembled backstage by an LLM that never appears as a participant. (B) Mean message toxicity in the fully logged treatment session dropped from 0.329 in the 20 minutes before the note to 0.220 in the 20 minutes after it, a 33% reduction (Welch $t(70.7) = 2.54, p = 0.013$).

Hostility between online communities is often rooted in group identity rather than substantive disagreement, and interventions that target what individuals believe leave that hostility largely untouched. Drawing on cooperative interdependence theory, we built a platform feature through which members of rival groups must co-author a single, publicly co-signed post: a Community Note with one contribution slot per group that publishes only when both sides have written their part. We deployed the feature on KFeed, a Reddit-style forum built for this research, in two-day sessions where members of ARMY and BLINK—the rival fandoms of BTS and BLACKPINK—discussed K-pop topics under persistent fandom badges, with a large language model monitoring toxicity and publishing the finished note entirely backstage. In the fully logged treatment session, mean message toxicity fell from 0.329 in the 20 minutes before the note to 0.220 in the 20 minutes after it, a 33% reduction; 4 of 5 participants who posted in both phases became less toxic, while survey measures of outgroup attitudes did not move. The most toxic participant did not soften even as the group did, and third-person references to the rival fandom gave way to first-person plural framing. We argue that the note works by interrupting the attention that escalation requires, not by producing a common identity, and discuss implications for designing conflict interventions that need not be accepted to work.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing**.

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1 INTRODUCTION

People like to fight online over who they are, not what to do. This hostility toward out-groups, rooted in group identity rather than substantive disagreement, is known as affective polarization [16, 17], and it pervades online communities [29, 36]. Common intuitive solutions target what people think. However, neither works reliably: exposure to the other side can entrench positions [2], and belief-correcting interventions produce modest, short-lived effects [38]. If the animosity is anchored in identity, interventions aimed at information miss the layer where the conflict lives.

Classic intergroup research suggests that hostility is reduced not by changing beliefs, but by changing relationships between groups. When rival groups become mutually dependent on a shared goal, cooperation fosters a common identity and reduces conflict [12, 30]. This mechanism, cooperative interdependence, has shown robust effects in real-world settings [21, 26]. However, existing online depolarization interventions target individuals rather than group relationships [8, 14, 41], and none brings rival groups into cooperative production. In this study, we investigate whether structured co-creation between rival groups can reduce intergroup hostility on online platforms.

We instantiate this idea as Community Notes, a platform feature through which members of rival groups co-author a single, publicly co-signed post. A pinned prompt offers one slot per group; the note publishes only when both sides have written their part, each entry verbatim under its author’s badge. Neither side can complete a note alone, and neither has to stop being itself. The LLM behind the feature stays out of sight: it monitors the discussion and publishes the finished note. Participants encounter a platform feature and each other, not a machine.

Testing such a feature requires rival groups whose hostility is real, identity-driven, and native to platforms. We chose K-pop fandom, one of the largest transnational fan cultures online [19, 24], and the rivalry between ARMY and BLINK, fandoms of BTS and BLACKPINK: it is enduring, conducted almost entirely on platforms, and rests on no substantive dispute, a near-pure case of identity-driven hostility [10, 42]. We deploy the feature on KFeed, a Reddit-style forum built for this research, where mixed sessions of ARMY and BLINK members discuss K-pop topics with or without the feature. We measure toxicity through automated scoring and human coding, and qualitatively analyze how cross-group interaction changes. We ask:

- **RQ1:** How do Community Notes affect polarized groups posting on online forums?
- **RQ2:** How does what they post change after the Community Note intervention?

In answering these questions, our work contributes: (1) a reusable platform mechanism that operationalizes cooperative interdependence in an online forum; (2) an extension of LLM-based depolarization from persuading individuals to structuring how groups interact; and (3) empirical evidence from real polarized communities on whether joint creation reduces toxicity.

2 RELATED WORK

Defusing polarization in online forums raises three questions: what kind of polarization it is, what reduces it, and what role an AI system can play. We first characterize affective polarization as a group-level phenomenon rooted in social identity. We then draw on cooperative interdependence theory to elicit the requirements for our intervention. Finally, we examine how LLMs have been used to intervene in online communities.

2.1 Affective Polarization in Online Fan Communities

Affective polarization refers to animosity toward an out-group that is rooted in group membership rather than substantive disagreement [16, 17]. Social identity theory traces this animosity to the self-concept: because people define themselves partly through their groups, favoring the in-group and derogating the out-group becomes a way of maintaining a positive self-image [34]. Strikingly, this requires almost nothing. In minimal group experiments, assigning people to arbitrary, meaningless categories was enough to produce discrimination between groups [33]. Hostility follows from the group boundary itself, not from any dispute across it. The intuitive remedy, correcting what each side believes about the other, therefore addresses the wrong layer. Polarization is a membership problem, not an information problem.

Political contexts make this hard to see, because partisan identity is entangled with real ideological stakes and the contribution of identity alone cannot be isolated [23]. Online fan communities present the phenomenon in purer form. Rival fandoms sustain hostility toward one another with no substantive dispute to resolve [10]. The bond runs deeper than preference: fans incorporate their idol’s successes and failures into their own self-worth, basking in reflected glory when the idol prevails [5], and the more strongly they identify, the less able they are to distance themselves when the idol is attacked [39].

This self-investment explains why fan conflict takes the form of costly, active aggression rather than mere dislike. Among the most devoted fans, personal and group identity can fuse, and fused individuals do not simply favor their group: they act for it, willingly bearing personal costs to defend it [31, 32]. Notably, this conflict unfolds within platform mechanics: fans fight with posts, downvotes, and coordinated campaigns, in forums and comment sections, at times over the use of those spaces themselves [22, 42]. Perceived threats to the group’s standing harden identification further rather than loosening it [3]. In idol and sports forums, toxicity and hostile language accordingly concentrate around encounters with rival groups [6, 29]. K-pop fandoms are a prominent case, organized, transnational, and highly active online [19, 24].

HCI has largely overlooked this side of fandom: the field’s fan research centers on positive collective action [11], and fandom itself remains a self-acknowledged gap [11]. Fan communities thus offer what political settings cannot: a near-pure case of identity-driven animosity, unfolding entirely within platform mechanics, and a clean setting for testing interventions that target identity rather than information.

2.2 Cooperative Interdependence and Intergroup Contact

The intuitive remedies for intergroup hostility are more contact and better information. Neither reliably changes identity-based conflict. Exposing partisans to opposing views entrenched their positions rather than softening them [2], and a megastudy of depolarization interventions found even the strongest effects modest and short-lived [38]. Yet one apparent failure reveals a design opportunity. In a field experiment, Iraqi Christians were assigned to amateur soccer teams that were either all-Christian or mixed with Muslim teammates [26]. Playing together did not change their general attitudes toward Muslims, but it changed their behavior. Months later, players from mixed teams were

more likely to train with Muslim peers and sign up for mixed leagues. Attitudes rooted in identity are difficult to shift directly, while cooperative behavior emerges when groups share a goal that neither side can achieve alone.

This mechanism is known as cooperative interdependence. In the Robbers Cave experiments, contact alone intensified competition between rival groups, until they were placed in situations requiring collaboration toward superordinate goals [30]. The Common Ingroup Identity Model explains why. Working toward a shared outcome leads people to recategorize themselves under a broader identity, extending the positive regard reserved for the in-group to former out-group members [12]. The mechanism operates among real-world rivals. Manchester United fans who encountered an injured jogger were more likely to help when he wore a United shirt than a Liverpool one; when a shared identity as football fans had been made salient beforehand, they helped the Liverpool supporter just as readily [21]. In both cases, behavior changed while group identities stayed intact.

These findings set three requirements for interventions targeting identity-driven conflict. Interaction must involve genuine interdependence, since tasks that either group can complete alone provide no motivation for recategorization [12]. The shared identity must be layered on top of existing subgroup identities rather than replacing them, because attempts to erase group boundaries read as identity threats, which intensify rather than reduce identification [3, 15]. And the shared identity must emerge from the interaction itself rather than being imposed explicitly, because categorization follows the identities made salient by the immediate situation [21, 37]. Our intervention is built to satisfy all three.

Online platforms are deeply involved in shaping these interactions. Platform affordances enable rival communities to organize, amplify, and sustain hostility at scale [22], and platform design can further intensify rivalry among fandoms [42]. But the same mechanisms that structure conflict can be redesigned to support cooperation. Existing approaches intervene at three points, setting aside moderation, which suppresses hostile expression without changing the relationship between groups. Exposure-based designs modify what users encounter across group boundaries, such as cross-cutting feeds or shared casual games [2, 14]. Matching-based designs modify who interacts with whom, pairing individuals across the divide [8]. Correction-based designs modify how contested information is evaluated, most notably Community Notes, where a bridging mechanism surfaces contributions considered helpful by users with otherwise opposing viewpoints [41].

These approaches leave the relationship between the groups unchanged. Polarized communities remain audiences exposed to one another, strangers temporarily paired, or adversaries evaluating each other's claims. None intervenes at the point where relationships are formed through joint creation, where opposing groups must produce something together rather than merely encounter, discuss, or judge one another. Cooperative interdependence, the mechanism with the strongest field evidence for changing intergroup behavior, has never been built into a platform and tested with members of real polarized communities.

2.3 LLM Interventions in Online Communities

When AI systems intervene in online conflict, they take one of the roles that conflict resolution research assigns to human third parties. Moderators enforce conduct rules after violations occur, facilitators manage participation without touching content, and mediators contribute questions and proposals of their own [25]. LLMs have now filled each seat. They moderate at scale [13], suggest real-time rephrasings that make contentious conversations feel heard [1], reduce belief in conspiracy theories through persuasive dialogue [9], and mediate live debates with psychologically grounded strategies [35]. Across this spectrum, what varies is how much the system contributes to the conversation.

Two features hold constant. First, the intervention targets individual minds. Each system persuades, corrects, or calms particular interlocutors, treating polarization as something wrong in what a person thinks. The results echo a

familiar pattern. LLM dialogue can move issue positions, yet affective polarization stays where it was [9, 35], the same dissociation Mousa observed on the soccer field, and the one expected if hostility follows membership rather than belief. Second, the system is a visible party to the exchange. Users know they are talking with, or being coached by, a machine, and disclosed AI authorship invites discounting and reactance [18]. The deeper problem is theoretical. An agent that openly steers people toward common ground is doing precisely what the third requirement rules out, announcing a shared identity rather than letting the situation produce one.

Across this spectrum, then, the LLM holds a seat in the interaction, and users are acted upon by the machine. No existing work inverts that allocation, returning the interaction entirely to the users while confining the system to the conditions under which they meet. In that position the LLM shapes when cooperation becomes possible and how a joint product takes form, but what participants encounter, throughout, is each other.

Together, the three literatures leave a single gap. Affective polarization has been studied almost entirely in political settings, leaving its purest identity-driven form in fan communities unexamined [10, 11]. Cooperative interdependence, the mechanism with the strongest field evidence, has never been built into a platform [26, 30]. And LLM interventions have varied the system’s role within the conversation without asking whether it belongs in the conversation at all [1, 35]. Our work sits at this intersection. We build a community notes feature through which members of rival fandoms co-author a single, publicly co-signed post, with the LLM working only backstage, and test it with two polarized communities on a live forum.

3 EXPERIMENT: COMMUNITY NOTES IN A POLARIZED FORUM

We conducted a two-day experiment to test whether a co-authorship feature reduces toxicity between rival fandoms. Participants, self-identified members of ARMY or BLINK, joined mixed sessions of ARMY and BLINK members on KFeed, a Reddit-style forum built for this research, and discussed K-pop topics under their fandom badge. Discussion threads were seeded with prompts, including toxic arguments drawn from pilot sessions, so that hostile, neutral, or constructive interaction could emerge naturally.

Throughout each session, the platform scored incoming posts for toxicity in real time. When session-level toxicity crossed a threshold determined from pilot sessions, the assigned feature appeared in the feed. We randomly assigned sessions to two conditions:

- (1) **Community Notes treatment:** the feature that appeared was a Community Note, a pinned prompt framed at the K-pop level with one contribution slot per fandom. The note published only when both sides had written their part, each entry preserved verbatim under its author’s badge.
- (2) **Inert feature control:** the feature that appeared was matched to Community Notes on visual salience and interaction cost but required no cross-fandom contribution. Its specific design is being finalized. Following the placebo-activity logic of Combs et al. [8], this condition ensures that both arms experience a triggered platform event, so that novelty and regression toward the mean are shared across conditions and the treatment effect is identified by the condition-by-phase contrast.

On Day 2, participants returned to the same session composition and discussed without any feature, allowing us to examine whether behavioral changes persisted after the mechanism was removed.

3.1 Building KFeed and the Community Notes Feature

KFeed supports community feeds, threaded discussion, moderator-pinned posts, and persistent fandom badges on every post. The Community Notes feature translates the three requirements from Section 2.2 into interface rules (Figure 1A).

- **A note needs both fandoms.** After one side submits, the entry sits waiting under its author’s badge until a rival answers. Neither side can write for the other or publish alone.
- **Both badges stay visible.** From draft to publication, each entry carries its author’s fandom badge, and votes accrue to the note as a whole rather than to either half.
- **Nothing is said about shared identity.** Prompts address K-pop fans at large, but no instruction tells participants to identify as one group, and entries are published verbatim rather than merged.

3.2 Configuring the Backstage LLM

An LLM performs two functions in the system, both invisible to participants. First, it monitors the discussion: incoming posts are scored for toxicity in real time, producing the session-level signal that triggers the assigned feature. Second, it publishes the Community Note: once both fandom slots are filled, it verifies that the note contains one entry per fandom and assembles the two entries verbatim into the pinned format. The model does not write or rewrite anyone’s words, does not converse with participants, and claims no identity. We keep it backstage deliberately: an agent that visibly steered rivals toward common ground would announce the shared identity that the design requires the situation to produce. In our system, the LLM decides when the cooperative task appears and makes its outcome public; the cooperation itself belongs entirely to the participants.

3.3 Recruitment and Participants

We recruited participants through fandom communities on social platforms, screening with an open-ended fandom identification question to avoid signaling the study’s focus, with working language and location as eligibility criteria.

3.4 Procedure

Participants completed a pre-survey at recruitment (fandom identification, trait covariates, and eligibility screening), then joined fixed time slots and confirmed readiness before each session. Sessions lasted approximately one hour per day on two consecutive days. A survey followed each day’s session: after Day 1, the outcome battery; after Day 2, the outcome battery again, feature-specific items, and a funneled suspicion probe. After the Day 2 survey, we invited a subset of participants to semi-structured interviews about their experience across both days, sampling from both fandoms and both conditions.

3.5 Measures

Toxicity is the primary outcome, scored on pre-moderation text by Perspective API [20] and independent human coding following Coe, Kenski, and Rains’s incivility codebook [7], supplemented by a K-pop incivility lexicon validated before use with inter-rater reliability reported. Survey outcomes adapt validated instruments to the fandom boundary, following the battery of Rajadesingan et al. [28]: the feeling thermometer toward rival-fandom members [17], common in-group identity representation as the mediator [12], and fandom identification via a fan identification scale [40], which also tests that the intervention does not erode subgroup identity. Behavioral logs capture cross-fandom replies, votes, and we/they pronoun ratios [27]. Reactance serves as a backfire guardrail [4].

Table 1. The studied session by phase: free discussion (first 20 minutes) versus the note phase (after the Community Note appeared). Toxicity fell while first-person plural framing replaced third-person references to the rival fandom.

	Free phase	Note phase
Messages	39	42
Mean toxicity	0.329	0.220
“They” references per message	0.41	0.21
“We” references per message	0.08	0.10
“They” share of plural references	0.842	0.692

3.6 Analysis

For RQ1, we compare toxicity across conditions and phases; because both conditions are triggered at the same threshold, the treatment effect is identified by the condition-by-phase contrast rather than by a before-after comparison, and Day 2 tests whether changes persist. For RQ2, two researchers will code forum posts and interview transcripts for shifts in cross-group interaction.

4 RESULTS

We have run 5 two-day sessions to date, each mixing ARMY and BLINK members who discussed K-pop topics under a prominent badge marking their fandom. On Day 1, participants discussed freely until session toxicity reached a threshold, which triggered Community Notes; on Day 2, the same participants discussed without it. The behavioral analyses below draw on the treatment session with complete message-level logs: 6 participants (three ARMY, three BLINK, anonymized A1–A3 and B1–B3) exchanged 81 messages over 48 minutes on Day 1, with the Community Note appearing 20 minutes into the session. Day-level comparisons additionally use the platform’s per-participant session summaries across treatment sessions, and survey results draw on the 10 treatment-arm participants who completed the post-session battery. Message-level toxicity is scored 0–1.

4.1 Behavior Moved, Attitudes Did Not (RQ1)

4.1.1 Toxicity fell when the note appeared. Toxicity was high through the free phase, then dropped when Community Notes appeared. Mean message toxicity decreased from 0.329 to 0.220, a 33% reduction (Welch $t(70.7) = 2.54$, $p = 0.013$; Table 1).

Figure 2 shows the change in toxicity over ten-minute windows. Toxicity peaked at 0.355 in the window before the Community Note appeared. After the note appeared, every ten-minute window had lower toxicity than every window before it, falling to 0.117 in the final window. At the participant level, we compared the participants who posted in both phases; their mean within-participant difference was $M_{diff} = -0.108$ (paired $t(4) = -2.97$, $p = 0.041$), with $d_z = 1.33$.

4.1.2 Attitudes did not change. Participants rated the note low in legitimacy (2.20/5) and high in reactance (4.25/5). Perceived similarity to the rival fandom was near the scale midpoint (3.02/5), while the thermometer gap between their own and the rival fandom was 60 points (Table 2). These ratings suggest that participants did not respond to the note because they accepted its authority or felt more positively toward the rival fandom.

Survey responses and focus-group accounts pointed in different directions. While participants rated the note low in legitimacy, they described the pop-out itself as a useful feature: “*This function... well I don’t need to think about blinks*

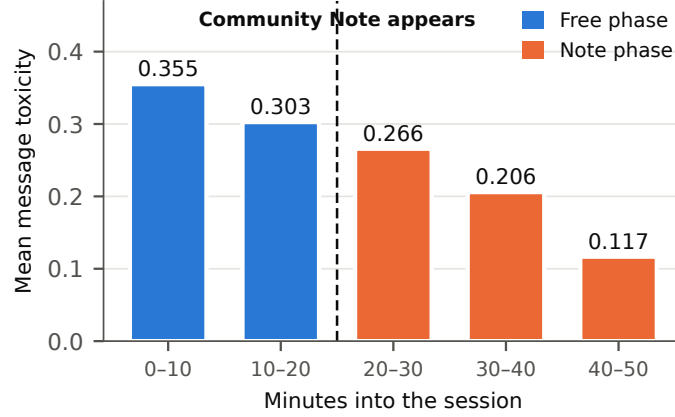


Fig. 2. Mean message toxicity in ten-minute windows across the studied session (n per window: 20, 19, 19, 17, 6 messages). The Community Note appeared 20 minutes in. Every window after the note is lower than every window before it.

Table 2. Post-session survey outcomes for treatment-arm participants ($n = 10$). Scale items are 1–5; thermometers are 0–100. Participants experienced the sessions as heated, granted the note little legitimacy, reported high reactance, and their outgroup attitudes stayed cold.

Measure	M (SD)
Legitimacy of the note	2.20 (0.89)
Reactance toward the note	4.25 (0.71)
Perceived similarity to rival fandom	3.02 (1.70)
Perception of rival fandom (intelligent, moral)	1.95 (0.90)
Cross-fandom contact intentions	3.02 (0.79)
Session felt heated / attacked	4.20 (0.79)
Thermometer: own fandom	79.8
Thermometer: rival fandom	19.7
Thermometer gap (own – rival)	60 (43.0)

too much. I just wrote my own opinion on the note and that's it. But at least it is useful because I...” (P#, focus group).
[TODO: quote is cut off; complete from transcript and replace P# with the participant code.]

4.2 The Effect Is Group-Level, Not Individual-Level (RQ1)

4.2.1 *The drop was not driven by a single participant.* The reduction in toxicity was spread across participants. Of the 5 participants who posted in both phases, 80% became less toxic (Figure 3). The one participant whose mean toxicity increased was among the least active in the session, making their individual mean less stable. One further participant posted only after the note appeared, at low toxicity.

The reduction remained when we removed participants one at a time, with the observed drop ranging from 0.078 to 0.135. Toxicity showed no significant trend within the free phase itself (slope -0.0076 per minute, $p = 0.20$), so the discontinuity at the note does not reflect a smooth pre-existing decline, though the short pre-period limits how strongly a trend can be ruled out.

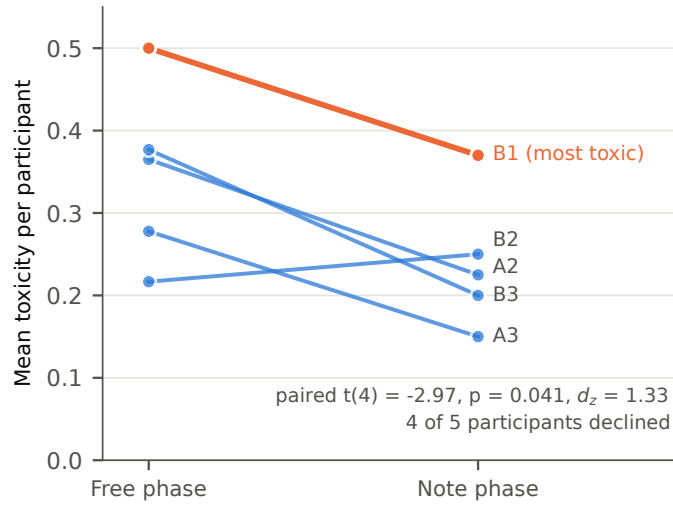


Fig. 3. Per-participant mean toxicity in the free phase versus the note phase, for the 5 participants who posted in both. 4 of 5 became less toxic; the most toxic participant (B1) declined least in relative terms and remained the most toxic voice in the room. Participant A1, who posted only after the note (4 messages, mean 0.175), is omitted.

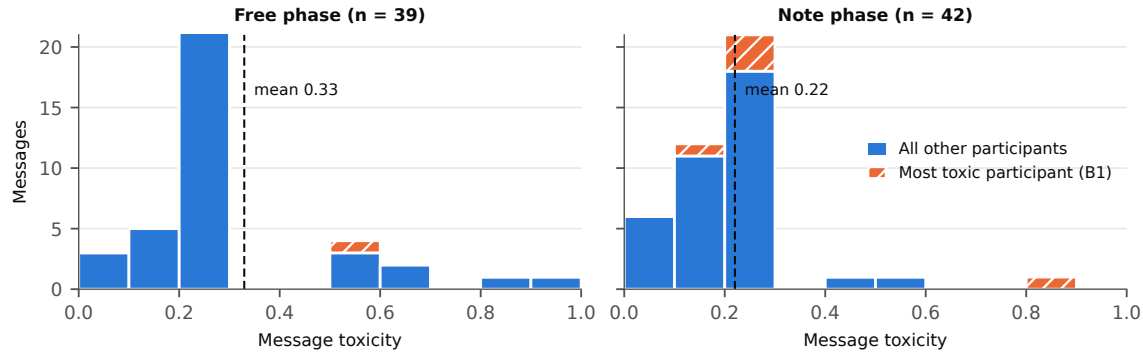


Fig. 4. Distribution of message toxicity before and after the Community Note appeared. The bulk of the distribution shifts left after the note, while the messages of the most toxic participant (B1, hatched) continue to occupy the right tail: the group softened; its most toxic member did not.

4.2.2 Hostility was concentrated from the start. A small number of participants produced much of the discussion. In the studied session, three of the six participants wrote 73% of the messages, and a single ARMY-BLINK pair (A2-B3) accounted for 17 of the session's 73 replies. These figures are from the single fully logged session; we will report the distribution across sessions as the remaining logs are processed.

The most active participant, B3, posted 13 messages during the free phase and 11 during the note phase, and their per-message toxicity fell from 0.377 to 0.200: the loudest voice softened with the group. The most toxic participant, B1, did not. Their note-phase per-message toxicity (0.370, with a message at 0.85) remained at the level of the whole group *before* the note, and the right tail of the note-phase toxicity distribution belongs almost entirely to them (Figure 4). The group distribution shifted left; its most toxic member stayed where they were.

4.3 The Topic Shift and the Toxicity Drop Occurred Together (RQ2)

4.3.1 *Conflict moved from works to persons to identity.* Focus-group participants described a similar escalation pattern across discussions. Conflicts often began with disagreements about works, such as comebacks, album sales, view and like counts on YouTube, or choreography. They then shifted to comments about the idol, followed by comments about the other participant, and finally to broader statements about fandom identity: “...when they [BLINKs] talk about BTS, they would say that oh, we ARMY have a fetish of ugly men. They would say we have a problem with aesthetics.” (P#, focus group). [TODO: replace P# with the participant code.]

4.3.2 *The note redirected the conversation rather than ending it.* Participants continued talking after the note appeared, but their attention shifted toward the note itself. In the focus groups, participants described the interruption as briefly taking them out of the argument, a state one described as *tuozhan*, or standing down: not agreement, but stepping away from that particular fight. [TODO: add supporting quote from transcript.]

A second measure showed a change at the same time. Third-person plural references fell from 0.41 to 0.21 per message, while first-person plural references increased from 0.08 to 0.10. The proportion of plural references using “they” therefore fell from 0.842 to 0.692 (Table 1).

Together, these results show that the conversation changed at the same time as toxicity decreased. We do not claim that the topic shift caused the reduction in toxicity.

4.4 The Effect Did Not Persist to Day 2

On Day 2 the same participants returned and discussed without the feature. Their mean toxicity (0.267 across 4 active participant-sessions) sat between the Day 1 free phase (0.347) and the Day 1 note phase (0.228), distinguishable from neither at these sample sizes (free vs. Day 2: Welch $t = 0.91$, $p = 0.40$; note vs. Day 2: Welch $t = -0.47$, $p = 0.66$; Figure 5). Across all treatment participant-sessions, Day 1 and Day 2 toxicity likewise did not differ (0.271 vs. 0.233; Welch $t = 0.54$, $p = 0.61$). The reduction observed while the note was on screen did not carry over once it was gone, a pattern we return to in Section 5.2.

5 DISCUSSION

5.1 A Shared Task Without a Shared Identity

The CIIM framework was necessary, but it worked differently from how we expected. It did not produce a common identity. It produced a shared task that both sides were willing to enter for a short time.

We built the feature to satisfy three requirements: real interdependence (neither side can publish alone), dual identities kept visible (badges remain), and no explicit common-identity framing (prompts address K-pop fans without asking anyone to identify as one group).

Behavior changed, but recategorization did not occur. Outgroup similarity stayed at the scale midpoint (3.02/5), the own-rival thermometer gap remained at 60 points, and perception of the rival fandom stayed low (1.95/5). Participants also rated the note low in legitimacy (2.20/5) and high in reactance (4.25/5). If the mechanism were “we now see them as one of us,” we would expect some loosening in outgroup attitudes or identification. We did not observe it.

The shared task was still doing work. If interruption alone were sufficient, any pop-up would serve: an announcement, a poll, an image. What we required instead was that each fandom write half of a note that could not publish until both halves existed. Without a superordinate task both sides are willing to enter, participants have nowhere to go: they either ignore the interruption or read it as one more position and absorb it into the argument.

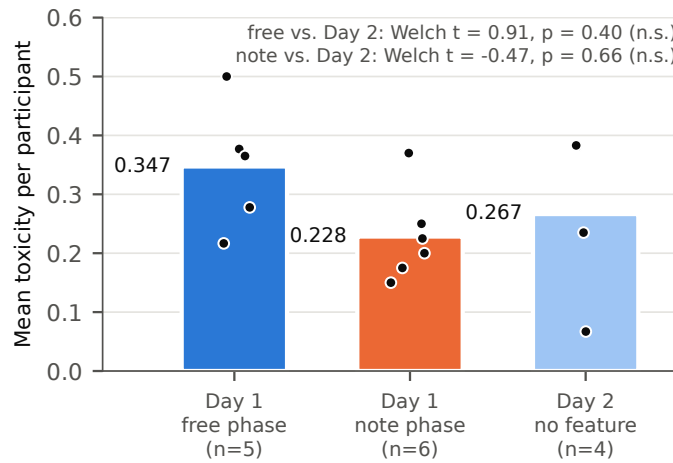


Fig. 5. Participant-level mean toxicity in the Day 1 free phase, the Day 1 note phase, and the Day 2 session without the feature (bars: group means; dots: participants). Day 2 sits between the two Day 1 phases and differs significantly from neither.

The common-identity framing and the pause are therefore layered rather than alternative. The framing supplies somewhere to go once the argument is left; the pause is the step that does the work, and it stops short of recategorization. Moving attention was enough.

This also accounts for low legitimacy alongside falling toxicity. Participants did not accept the note’s authority over their dispute, but they did accept the task of writing something as a K-pop fan. The objects of judgment differ: the shared identity supplied a reason to participate, not an outcome of identification.

One alternative reading should be kept. The shift from “they” to “we” looks like common identity, but a change of subject produces the same linguistic pattern. We treat it as parallel to the attention account rather than as evidence for CIIM.

This claim is testable. If interruption alone were doing the work, an inert feature matched on visual salience but requiring no cross-fandom contribution should produce the same effect. The control condition is a direct test of it.

5.2 Why a Pause Can Be Enough

Escalation requires sustained attention. In our seeded sessions toxicity did not need time to build—threads opened hot, and the free phase held a mean near 0.329—but it did need attention to stay hot. Escalation takes participants staying with the same fight, and once the note pulled attention elsewhere, no ten-minute window returned to pre-note levels; by the final window toxicity had fallen to 0.117.

Interrupting attention may therefore be enough to interrupt escalation, without changing what anyone believes. Depolarization interventions typically target beliefs, attitudes, group boundaries, or perceptions of the outgroup. Community Notes competes for none of these. It competes for where attention sits in the moment, which is a lower bar: participants need neither accept another position nor come to like the other side. But not every interruption will do. Attention needs somewhere to land—something both sides are willing to enter and must complete together. This is what the shared task provides. It does not change who participants are; it gives them somewhere to go for a few minutes.

Attention is itself scarce on platforms. Recommendation, reply, and engagement mechanics keep users inside conflicts [29], and participants themselves attributed sustained hostility to platform and industry incentives. An attention-based intervention need not compete with those mechanics over positions. It needs to win a few minutes.

Returning toward baseline on Day 2 (Figure 5) is therefore not evidence of failure. If the mechanism is attention interruption rather than attitude change, the effect should be state-dependent. The note does not buy reconciliation; it buys time. In an escalating conflict, an interrupted climb has to be rebuilt.

5.3 The Intervention Did Not Need to Be Accepted to Work

Toxicity fell alongside low legitimacy (2.20/5) and very high reactance (4.25/5). This is not a contradiction once the objects of resistance and participation are separated. Participants resisted the note’s standing to intervene in their argument. What they accepted was the task of writing something as a K-pop fan, which asked them neither to concede anything to the other side nor to give up their own position. High reactance attached to the note’s authority without fully translating into resistance to the behavioral intervention.

Prior work notes that visible AI invites discounting and reactance [18]. In our design the LLM never appeared as a persuader: participants encountered an interface feature rather than an agent telling them how to talk. This may be one reason resistance did not spread across the whole intervention.

The design implication is that an effective conflict intervention need not be endorsed by its users. Mechanisms that do not depend on goodwill or agreement may be easier to deploy than mechanisms that depend on persuasion, because they need not win consent first. This is consistent with the second point above: competing for attention has a lower bar than competing for belief.

We claim only that high reactance did not prevent the observed behavioral change. We cannot show that reactance is unimportant, nor rule out that higher reactance would defeat the intervention.

5.4 The Effect Reached the Periphery, Not the Core

Hostility was unevenly distributed. Three participants produced 73% of messages, and a single ARMY-BLINK pair carried 17 of 73 replies.

The most toxic participant is the key boundary case. While the group distribution shifted left after the note, their per-message toxicity barely moved: at 0.370 it remained where the whole group had been before the note, and the note-phase right tail of the toxicity distribution is almost entirely theirs (Figure 4). Notably, volume and venom came apart: the most active participant softened with the group, while the most toxic one kept their edge.

Community Notes therefore appears to soften the behavior of peripheral participants rather than to change the hostility of core producers. Group means fell while the participant most invested in the conflict did not visibly change.

The mechanism above explains this boundary. The pause works because a shared task is available for both sides to enter. For participants already heavily committed to the fight, that task is not an attractive place to go: their attention stays anchored to the argument. The shared task catches those still willing to leave.

For cooperative interdependence, this suggests that online cooperative mechanisms may reach those still willing to participate more readily than those already committed to the fight. This limits where the mechanism applies rather than undermining it.

The design implication is that triggering should not depend only on the state of the whole discussion. A better design would combine individual trajectories—who is escalating, who is being drawn in, who is moving from the periphery toward the core—shifting the trigger from a session-level threshold to a participant-level trajectory.

5.5 Limitations

Four limitations bound these results. First, the message-level analyses draw on the single treatment session with complete logs; the remaining sessions enter the analysis only through per-participant summaries, and the control condition is still being fielded, so the condition-by-phase contrast that identifies the treatment effect is not yet available. Second, the note onset in the analyzed export is reconstructed as the 20-minute mark rather than read from a trigger log, so the phase boundary carries some imprecision. Third, day-level comparisons treat platform participant-sessions as the unit of analysis; participants who reconnected contribute more than one row. Fourth, the survey battery analyzed here was collected after Day 1 only, so attitude stability is inferred from post-session levels rather than pre-post change. Each of these will be addressed as the full dataset is assembled.

6 CONCLUSION

We built a platform feature that operationalizes cooperative interdependence: a Community Note that members of rival fandoms must co-author before it can publish, assembled backstage by an LLM that never enters the conversation. In a live session between ARMY and BLINK members, toxicity fell by 33% when the note appeared, the decline was shared by most participants rather than driven by one, and conversation shifted from “they” toward “we”—while attitudes toward the rival fandom did not move and the note itself was granted little legitimacy. The pattern is consistent with an attention account: the co-authored note gives an escalating fight somewhere else to go, without asking anyone to change what they believe or who they are. A pause, it turns out, can be built.

REFERENCES

- [1] Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. 2023. Leveraging AI for Democratic Discourse: Chat Interventions Can Improve Online Political Conversations at Scale. *Proceedings of the National Academy of Sciences* 120, 41 (2023), e2311627120. <https://doi.org/10.1073/pnas.2311627120>
- [2] Christopher A. Bail, Lisa P. Argyle, Taylor W. Brown, John P. Bumpus, Haohan Chen, M. B. Fallin Hunzaker, Jaemin Lee, Marcus Mann, Friedolin Merhout, and Alexander Volfovsky. 2018. Exposure to Opposing Views on Social Media Can Increase Political Polarization. *Proceedings of the National Academy of Sciences* 115, 37 (2018), 9216–9221. <https://doi.org/10.1073/pnas.1804840115>
- [3] Nyla R. Branscombe, Naomi Ellemers, Russell Spears, and Bertjan Doojsje. 1999. The Context and Content of Social Identity Threat. In *Social Identity: Context, Commitment, Content*, Naomi Ellemers, Russell Spears, and Bertjan Doojsje (Eds.). Blackwell, Oxford, 35–58.
- [4] Jack W. Brehm. 1966. *A Theory of Psychological Reactance*. Academic Press, New York, NY.
- [5] Robert B. Cialdini, Richard J. Borden, Avril Thorne, Marcus R. Walker, Stephen Freeman, and Lloyd R. Sloan. 1976. Basking in Reflected Glory: Three (Football) Field Studies. *Journal of Personality and Social Psychology* 34, 3 (1976), 366–375. <https://doi.org/10.1037/0022-3514.34.3.366>
- [6] Jamie Cleland. 2014. Racism, Football Fans, and Online Message Boards: How Social Media Has Added a New Dimension to Racist Discourse in English Football. *Journal of Sport and Social Issues* 38, 5 (2014), 415–431. <https://doi.org/10.1177/0193723513499922>
- [7] Kevin Coe, Kate Kenski, and Stephen A. Rains. 2014. Online and Uncivil? Patterns and Determinants of Incivility in Newspaper Website Comments. *Journal of Communication* 64, 4 (2014), 658–679. <https://doi.org/10.1111/jcom.12104>
- [8] Aidan Combs, Graham Tierney, Brian Guay, Friedolin Merhout, Christopher A. Bail, D. Sunshine Hillygus, and Alexander Volfovsky. 2023. Reducing Political Polarization in the United States with a Mobile Chat Platform. *Nature Human Behaviour* 7 (2023), 1454–1461. <https://doi.org/10.1038/s41562-023-01655-0>
- [9] Thomas H. Costello, Gordon Pennycook, and David G. Rand. 2024. Durably Reducing Conspiracy Beliefs Through Dialogues with AI. *Science* 385, 6714 (2024), eadq1814. <https://doi.org/10.1126/science.adq1814>
- [10] Mark Duffett. 2013. *Understanding Fandom: An Introduction to the Study of Media Fan Culture*. Bloomsbury, New York, NY.
- [11] Casey Fiesler, Shannon Morrison, and Amy S. Bruckman. 2016. An Archive of Their Own: A Case Study of Feminist HCI and Values in Design. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, 2574–2585. <https://doi.org/10.1145/2858036.2858409>
- [12] Samuel L. Gaertner and John F. Dovidio. 2000. *Reducing Intergroup Bias: The Common Ingroup Identity Model*. Psychology Press, Philadelphia, PA.
- [13] Fabrizio Gilardi, Meysam Alizadeh, and Maël Kübli. 2023. ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks. *Proceedings of the National Academy of Sciences* 120, 30 (2023), e2305016120. <https://doi.org/10.1073/pnas.2305016120>

- [14] Andrew M. Guess, Neil Malhotra, Jennifer Pan, Pablo Barberá, Hunt Allcott, Taylor Brown, Adriana Crespo-Tenorio, Drew Dimmery, Deen Freelon, Matthew Gentzkow, et al. 2023. How Do Social Media Feed Algorithms Affect Attitudes and Behavior in an Election Campaign? *Science* 381, 6656 (2023), 398–404. <https://doi.org/10.1126/science.abp9364>
- [15] Matthew J. Hornsey and Michael A. Hogg. 2000. Assimilation and Diversity: An Integrative Model of Subgroup Relations. *Personality and Social Psychology Review* 4, 2 (2000), 143–156. https://doi.org/10.1207/S15327957PSPR0402_03
- [16] Shanto Iyengar, Yphtach Lelkes, Matthew Levendusky, Neil Malhotra, and Sean J. Westwood. 2019. The Origins and Consequences of Affective Polarization in the United States. *Annual Review of Political Science* 22 (2019), 129–146. <https://doi.org/10.1146/annurev-polisci-051117-073034>
- [17] Shanto Iyengar, Gaurav Sood, and Yphtach Lelkes. 2012. Affect, Not Ideology: A Social Identity Perspective on Polarization. *Public Opinion Quarterly* 76, 3 (2012), 405–431. <https://doi.org/10.1093/poq/nfs038>
- [18] Maurice Jakesch, Megan French, Xiao Ma, Jeffrey T. Hancock, and Mor Naaman. 2019. AI-Mediated Communication: How the Perception that Profile Text was Written by AI Affects Trustworthiness. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, 1–13. <https://doi.org/10.1145/3290605.3300469>
- [19] Dal Yong Jin and Kyong Yoon. 2016. The Social Mediascape of Transnational Korean Pop Culture: Hallyu 2.0 as Spectacle. *New Media & Society* 18, 7 (2016), 1277–1292. <https://doi.org/10.1177/1461444814554895>
- [20] Alyssa Lees, Vinh Q. Tran, Yi Tay, Jeffrey Sorensen, Jai Gupta, Donald Metzler, and Lucy Vasserman. 2022. A New Generation of Perspective API: Efficient Multilingual Character-Level Transformers. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, New York, NY, 3197–3207. <https://doi.org/10.1145/3534678.3539147>
- [21] Mark Levine, Amy Prosser, David Evans, and Stephen Reicher. 2005. Identity and Emergency Intervention: How Social Group Membership and Inclusiveness of Group Boundaries Shape Helping Behavior. *Personality and Social Psychology Bulletin* 31, 4 (2005), 443–453. <https://doi.org/10.1177/0146167204271651>
- [22] Alice E. Marwick. 2021. Morally Motivated Networked Harassment as Normative Reinforcement. *Social Media + Society* 7, 2 (2021). <https://doi.org/10.1177/20563051211021378>
- [23] Lilliana Mason. 2018. *Uncivil Agreement: How Politics Became Our Identity*. University of Chicago Press, Chicago, IL.
- [24] Courtney McLaren and Dal Yong Jin. 2020. “You Can’t Help But Love Them”: BTS, Transcultural Fandom, and Affective Identities. *Korea Journal* 60, 1 (2020), 100–127. <https://doi.org/10.25024/kj.2020.60.1.100>
- [25] Christopher W. Moore. 2014. *The Mediation Process: Practical Strategies for Resolving Conflict* (4th ed.). Jossey-Bass, San Francisco, CA.
- [26] Salma Mousa. 2020. Building Social Cohesion Between Christians and Muslims Through Soccer in Post-ISIS Iraq. *Science* 369, 6505 (2020), 866–870. <https://doi.org/10.1126/science.abb3153>
- [27] James W. Pennebaker. 2011. *The Secret Life of Pronouns: What Our Words Say About Us*. Bloomsbury Press, New York, NY.
- [28] Ashwin Rajadesingan, Paul Resnick, and Ceren Budak. 2020. Quick, Community-Specific Learning: How Distinctive Toxicity Norms Are Maintained in Political Subreddits. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 14. AAAI Press, Palo Alto, CA, 557–568. <https://doi.org/10.1609/icwsm.v14i1.7323>
- [29] Steve Rathje, Jay J. Van Bavel, and Sander van der Linden. 2021. Out-group Animosity Drives Engagement on Social Media. *Proceedings of the National Academy of Sciences* 118, 26 (2021), e2024292118. <https://doi.org/10.1073/pnas.2024292118>
- [30] Muzafer Sherif, O. J. Harvey, B. Jack White, William R. Hood, and Carolyn W. Sherif. 1961. Intergroup Conflict and Cooperation: The Robbers Cave Experiment. In *Intergroup Conflict and Cooperation: The Robbers Cave Experiment*. University Book Exchange, Norman, OK.
- [31] William B. Swann, Ángel Gómez, D. Conor Seyle, J. Francisco Morales, and Carmen Huici. 2009. Identity Fusion: The Interplay of Personal and Social Identities in Extreme Group Behavior. *Journal of Personality and Social Psychology* 96, 5 (2009), 995–1011. <https://doi.org/10.1037/a0013668>
- [32] William B. Swann, Jolanda Jetten, Ángel Gómez, Harvey Whitehouse, and Brock Bastian. 2012. When Group Membership Gets Personal: A Theory of Identity Fusion. *Psychological Review* 119, 3 (2012), 441–456. <https://doi.org/10.1037/a0028589>
- [33] Henri Tajfel, M. G. Billig, R. P. Bundy, and Claude Flament. 1971. Social Categorization and Intergroup Behaviour. *European Journal of Social Psychology* 1, 2 (1971), 149–178. <https://doi.org/10.1002/ejsp.2420010202>
- [34] Henri Tajfel and John C. Turner. 1979. An Integrative Theory of Intergroup Conflict. In *The Social Psychology of Intergroup Relations*, William G. Austin and Stephen Worchel (Eds.). Brooks/Cole, Monterey, CA, 33–47.
- [35] Michael Henry Tessler, Michiel A. Bakker, Daniel Jarrett, Hannah Sheahan, Martin J. Chadwick, Raphael Koster, Georgina Evans, Lucy Campbell-Gillingham, Tatum Collins, David C. Parkes, Matthew Botvinick, and Christopher Summerfield. 2024. AI Can Help Humans Find Common Ground in Democratic Deliberation. *Science* 386, 6719 (2024), eadq2852. <https://doi.org/10.1126/science.adq2852>
- [36] Petter Törnberg. 2022. How Digital Media Drive Affective Polarization Through Partisan Sorting. *Proceedings of the National Academy of Sciences* 119, 42 (2022), e2207159119. <https://doi.org/10.1073/pnas.2207159119>
- [37] John C. Turner, Michael A. Hogg, Penelope J. Oakes, Stephen D. Reicher, and Margaret S. Wetherell. 1987. *Rediscovering the Social Group: A Self-Categorization Theory*. Basil Blackwell, Oxford.
- [38] Jan G. Voelkel, Michael N. Stagnaro, James Y. Chu, Sophia L. Pink, Joseph S. Mernyk, Chrystal Redekopp, Isaias Ghezze, Matthew Cashman, Dhaval Adjodah, Levi G. Allen, et al. 2024. Megastudy Identifying Effective Interventions to Strengthen Americans’ Democratic Attitudes. *Science* 386, 6719 (2024), eadh4764. <https://doi.org/10.1126/science.adh4764>
- [39] Daniel L. Wann and Nyla R. Branscombe. 1990. Die-Hard and Fair-Weather Fans: Effects of Identification on BIRGing and CORFing Tendencies. *Journal of Sport and Social Issues* 14, 2 (1990), 103–117. <https://doi.org/10.1177/019372359001400203>

- [40] Daniel L. Wann and Nyla R. Branscombe. 1993. Sports Fans: Measuring Degree of Identification with Their Team. *International Journal of Sport Psychology* 24, 1 (1993), 1–17.
- [41] Stefan Wojcik, Sophie Hilgard, Nick Judd, Delia Mocanu, Stephen Ragain, M. B. Fallin Hunzaker, Keith Coleman, and Jay Baxter. 2022. Birdwatch: Crowd Wisdom and Bridging Algorithms Can Inform Understanding and Reduce the Spread of Misinformation. *arXiv preprint arXiv:2210.15723* (2022). <https://doi.org/10.48550/arXiv.2210.15723>
- [42] Yiyi Yin. 2020. An Emergent Algorithmic Culture: The Data-ization of Online Fandom in China. *International Journal of Cultural Studies* 23, 4 (2020), 475–492. <https://doi.org/10.1177/1367877920908269>